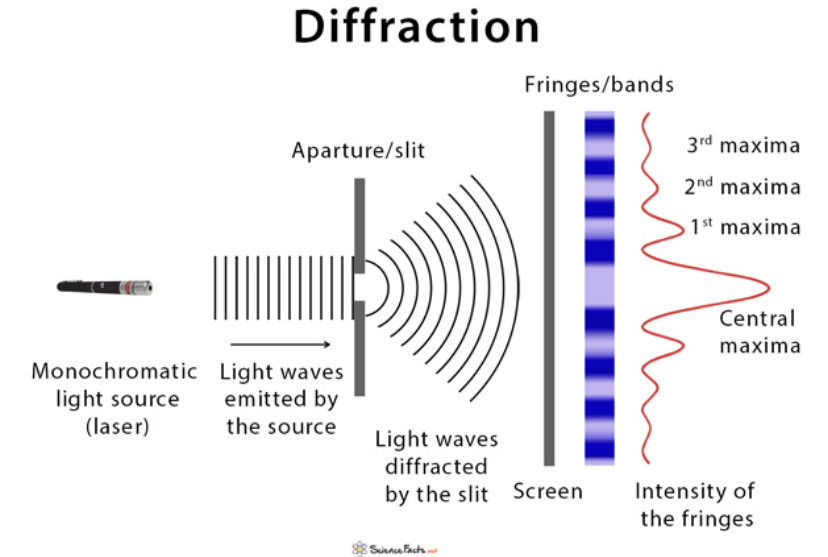
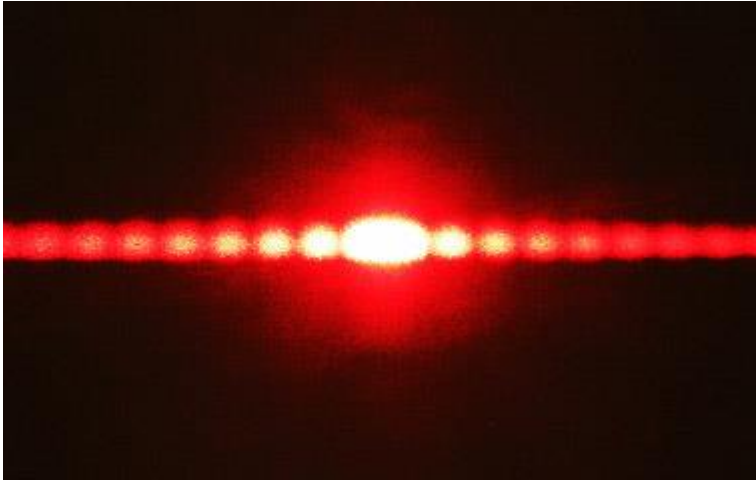


Wave and Optics

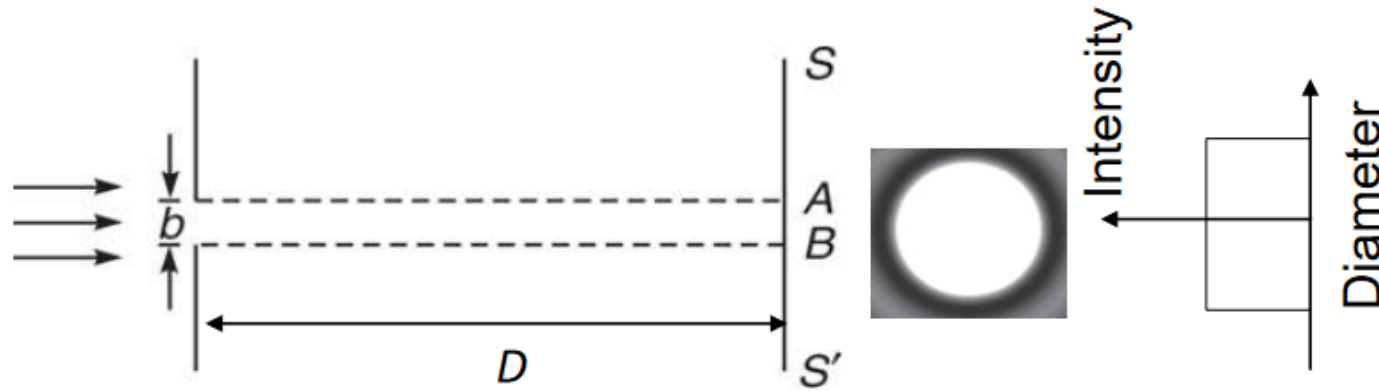
Part-4



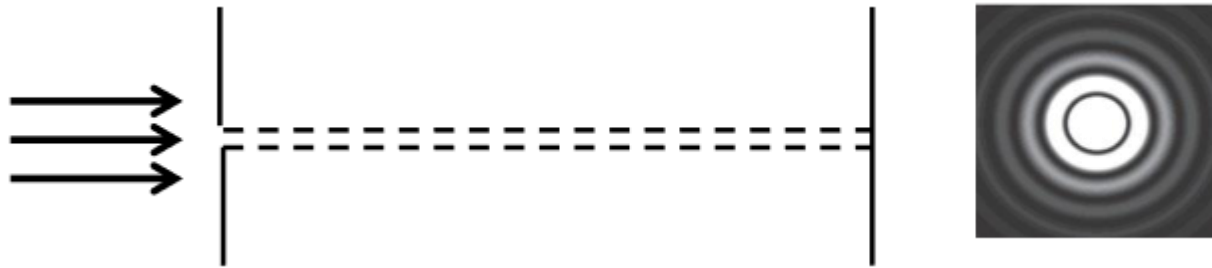
Dr. Ram Prakash
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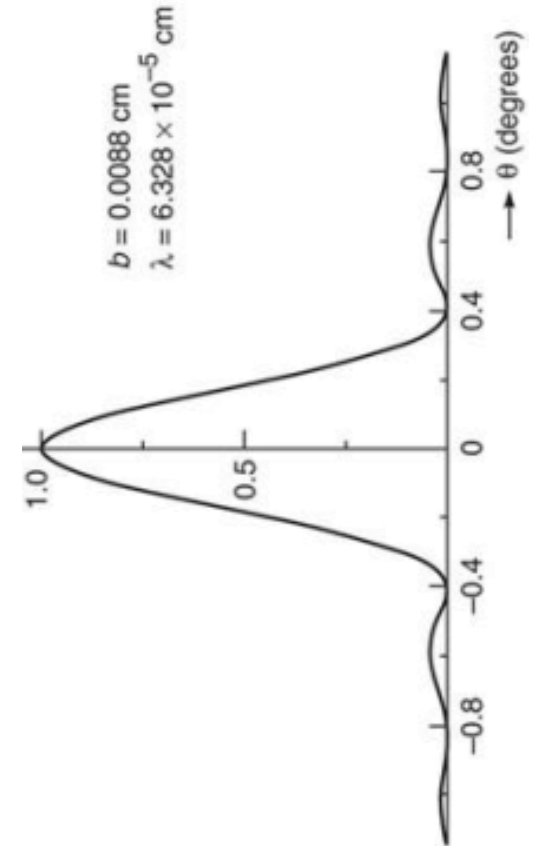
- A plane wave of wavelength λ incident on a circular aperture of diameter $2b$ ($\gg \lambda$) on a screen and a screen is placed at a distance D behind the aperture what will you observe?



- If size b of the aperture is significantly reduced to few times λ i.e. comparable to λ



$\Rightarrow$ Diffraction due to circular aperture



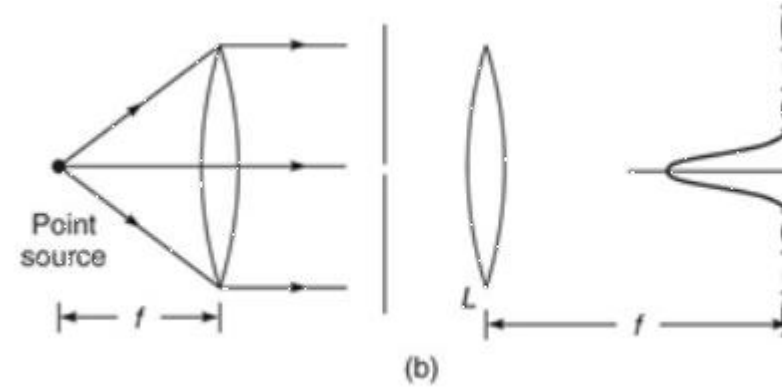
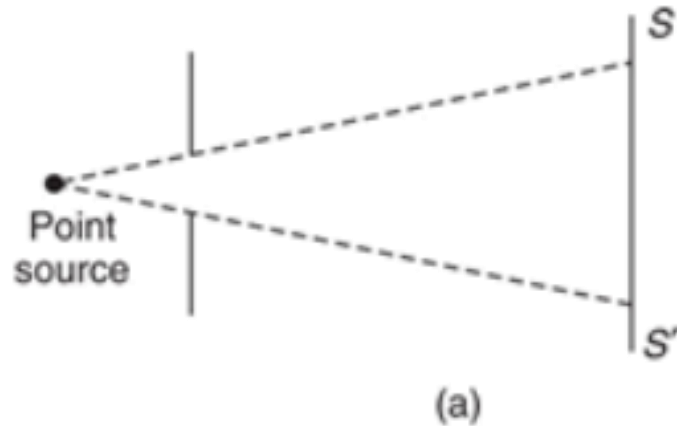
Diffraction

- When wave encounter an edge or an obstacle or an aperture with a size comparable to the wavelength of the waves, these waves spread out as they travel and as a result undergo interference. This phenomena is called as Diffraction.
- One finds that the spreading out decreases with decrease in wavelength. So, more spreading means more diffraction.
- Since the wavelengths of light are very small, the effects due to diffraction are not readily observed.
- Wavelengths of sounds are much larger as compared to light, so diffraction in sound waves are quite evident in our daily experiences.

- 1835: George Airy calculated diffraction pattern due to a circular aperture
- 1819: Joseph Fraunhofer demonstrated diffraction by a grating and published his theory in 1823

- **Two classes of diffraction:**

a) **Fresnel diffraction:** Source and observation screen are at a finite distances from the aperture.



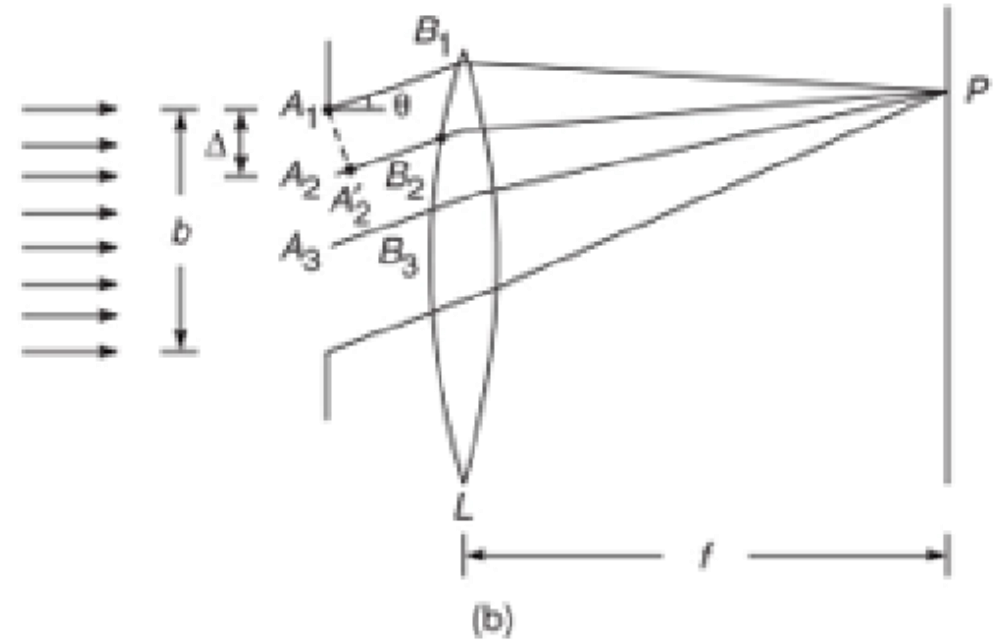
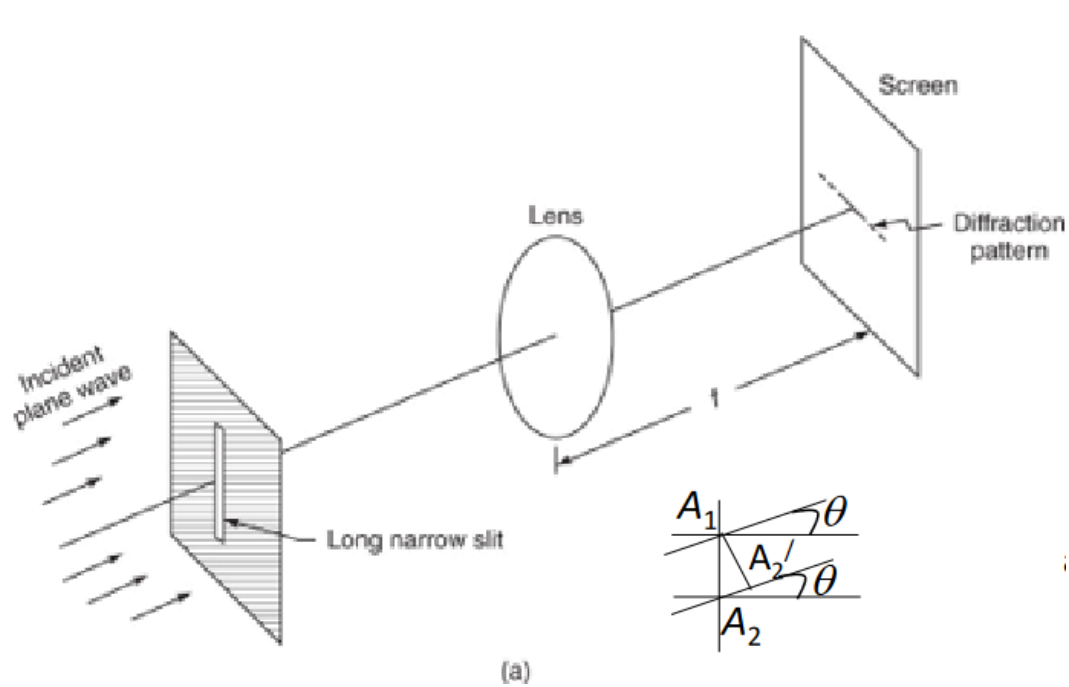
b) **Fraunhofer diffraction:** Source and observation screen are at infinite distances from the aperture.

How to realize
infinite
distance within
a laboratory?

Use a pair of convex lenses or a laser light, which emits almost parallel light.

❖ Fraunhofer diffraction is much easier to study/ model as compared to Fresnel diffraction.

a) Fraunhofer diffraction by a single slit:



Let the slit width be b . To obtain the intensity at P at the focal plane of Lens, assume the slit consists of N number of equally spaced (Δ) point sources $A_1, A_2, A_3, \dots$.

Fields due to sources $A_1, A_2, A_3, \dots$ will superimpose at P with different phases due to slight path length difference A_2A_2' between consecutive point sources.

If we assume N number of point sources

$$\Rightarrow b = (N - 1) \Delta$$

a) Fraunhofer diffraction by a single slit (contd...)

$$\angle C'A_1D' = \theta = \angle CA_1D$$

$$\angle CA_1A_2 = \frac{\pi}{2} - \theta = \angle A_1A_2A_2'$$

A_2' is foot of the normal from A_1 on $A_2'B_2$

Again,

$$\angle A_1A_2A_2' + \angle A_2A_1A_2' = \frac{\pi}{2} = \frac{\pi}{2} - \theta + \angle A_2A_1A_2'$$

$$\Rightarrow \angle A_2A_1A_2' = \theta$$

$$\Rightarrow A_2A_2' / \Delta = \sin \theta \quad \text{or} \quad A_2A_2' = \Delta \sin \theta$$

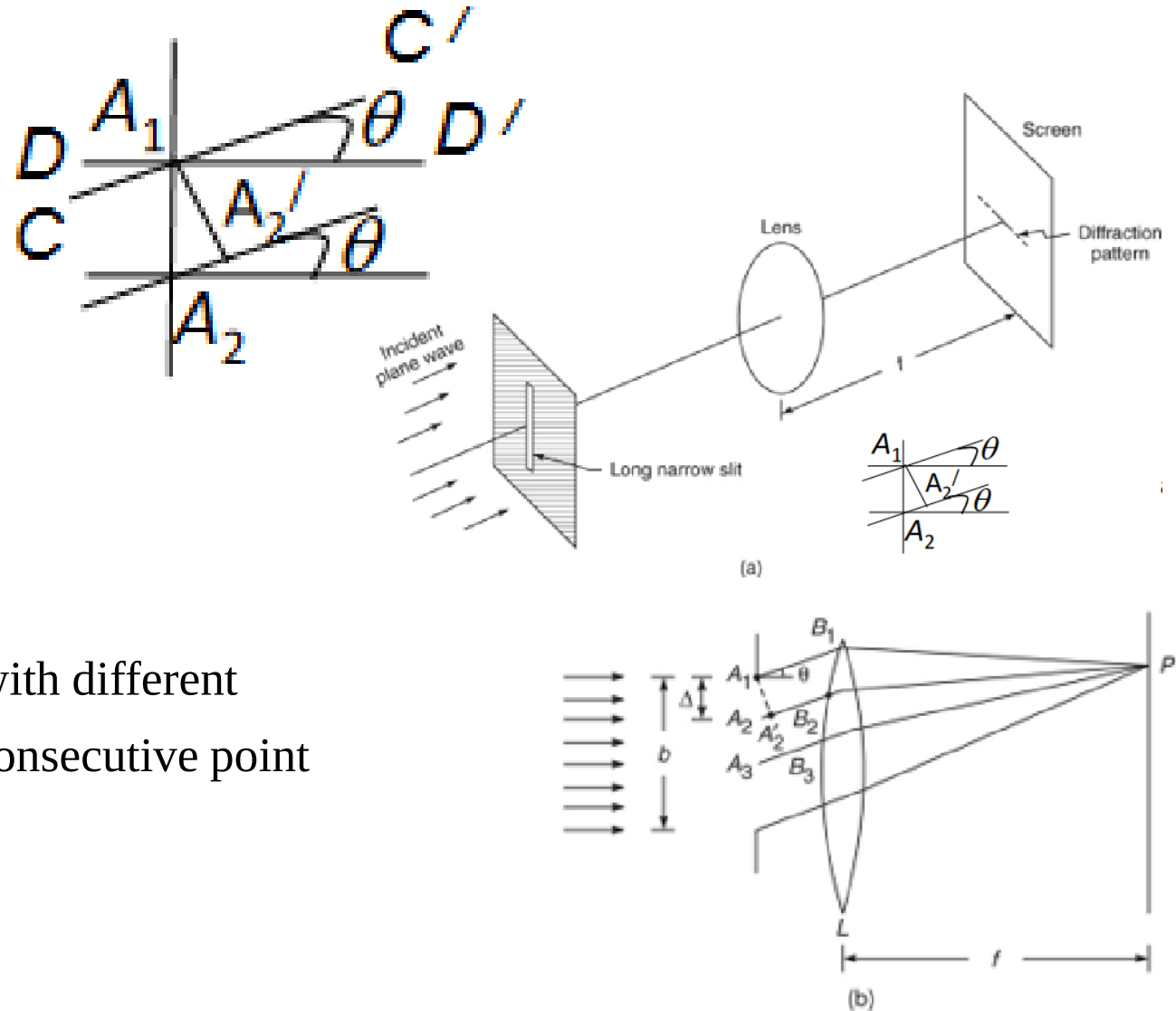
$$\Rightarrow \text{Corresponding phase diff: } \phi = \frac{2\pi}{\lambda_0} \Delta \sin \theta$$

Fields due to sources $A_{1,2,3,\dots}$ will superimpose at P with different phases due to slight path length difference between consecutive point sources.

Since we assumed N number of point sources

$$\Rightarrow b = (N - 1) \Delta$$

Likewise, subsequent rays will also be differing in phase from the previous one – all of which will superimpose at P.



The resultant electric field of the light at P will be

$$E = a \cos \omega t + a \cos(\omega t - \phi) + a \cos(\omega t - 2\phi) + \dots + \cos(\omega t - (N-1)\phi)$$

$$E = a [\cos \omega t + \cos(\omega t - \phi) + \cos(\omega t - 2\phi) + \dots + \cos(\omega t - (N-1)\phi)]$$

From complex analysis one can represent

$$E_1 = a \cos \omega t \text{ as real part of } E_1 = a e^{i\omega t}$$

Likewise

$$E_2 = a \cos(\omega t - \phi_1) \text{ is the real part of } a e^{i(\omega t - \phi_1)}$$

Thus one can express total E at P as

$$\begin{aligned} E &= E_1 + E_2 + E_3 + \dots = a e^{i\omega t} [1 + e^{-i\phi} + e^{-2i\phi} + \dots + e^{-i(N-1)\phi}] \\ &= a e^{i\omega t} \frac{1 - e^{-iN\phi}}{1 - e^{-i\phi}} = a e^{i\omega t} \frac{e^{-iN\phi/2}}{e^{-i\phi/2}} \times \frac{e^{+iN\phi/2} - e^{-iN\phi/2}}{e^{+i\phi/2} - e^{-i\phi/2}} \\ &= a \frac{e^{i[\omega t - N\phi/2]}}{e^{-i\phi/2}} \times \frac{\sin \frac{N\phi}{2}}{\sin \frac{\phi}{2}} = a \frac{\sin \frac{N\phi}{2}}{\sin \frac{\phi}{2}} \times e^{i[\omega t - (N-1)\phi/2]} \end{aligned}$$

Thus

$$E = E_{\theta} \cos[\omega t - (N-1)\phi/2]; \quad E_{\theta} = a \frac{\sin \frac{N\phi}{2}}{\sin \frac{\phi}{2}} \dots \dots \dots \quad (x)$$

Since $b = (N-1)\Delta$ In the limit, $N \rightarrow \infty$ and $\Delta \rightarrow 0 \Rightarrow N\Delta \rightarrow b$

$$\frac{N\phi}{2} = \frac{N}{2} \times \frac{2\pi}{\lambda_0} \Delta \sin \theta = \frac{\pi}{\lambda_0} N\Delta \sin \theta = \frac{\pi}{\lambda_0} b \sin \theta$$

Moreover

$$\phi = \frac{2\pi}{\lambda_0} \Delta \sin \theta = \frac{2\pi}{\lambda_0} \times \frac{b}{N} \times \sin \theta \quad \text{For large } N, \phi \rightarrow 0 \text{ and } \sin \frac{\phi}{2} \approx \frac{\phi}{2}$$

Hence

$$E_{\theta} \approx a \frac{\sin \frac{N\phi}{2}}{\frac{\phi}{2}} = a \frac{\sin (\frac{\pi}{\lambda_0} b \sin \theta)}{\frac{\pi}{N\lambda_0} b \sin \theta} = aN \frac{\sin (\frac{\pi}{\lambda_0} b \sin \theta)}{\frac{\pi}{\lambda_0} b \sin \theta} = A \frac{\sin \beta}{\beta}; \beta = \frac{\pi}{\lambda_0} b \sin \theta$$

$$\Rightarrow \text{From (x)} \quad E = A \frac{\sin \beta}{\beta} \cos (\omega t - \beta) \Rightarrow \text{Intensity : } |E|^2 = I = \left| A \frac{\sin \beta}{\beta} e^{i(\omega t - \beta)} \right|^2 = I_0 \frac{\sin^2 \beta}{\beta^2}$$



$$(N-1)\frac{\phi}{2} \approx N\frac{\phi}{2} = \frac{\pi}{\lambda_0} b \sin \theta = \beta$$

From

$$I = I_0 \frac{\sin^2 \beta}{\beta^2} \text{ for single-slit diffraction pattern}$$

- I will be minimum for $\beta = m\pi$; $m = 1, 2, 3, \dots$

Why is $m = 0$
not admissible
???

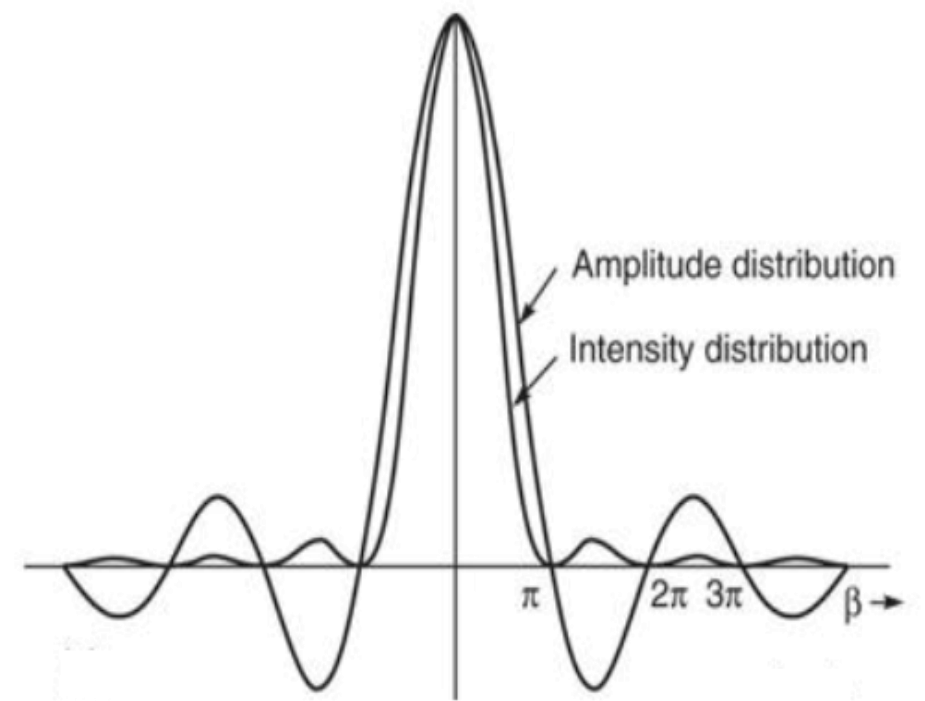
$$\text{Since } \frac{\sin(\beta=0)}{(\beta=0)} = 1 \Rightarrow I = I_0 \Rightarrow \text{maximum}$$

$$\Rightarrow \beta = \frac{\pi}{\lambda} b \sin \theta = m\pi; m = 1, 2, 3, \dots$$

$$\Rightarrow b \sin \theta = m\lambda; m = \pm 1, \pm 2, \pm 3, \dots$$

represents condition for minima

$$\Rightarrow 1^{\text{st}} \text{ minimum appears at } \theta = \sin^{-1}\left(\frac{\lambda}{b}\right), \quad 2^{\text{nd}} \text{ minimum appears at } \theta = \sin^{-1}\left(\frac{2\lambda}{b}\right); \dots$$



For maxima, we differentiate $I = I_0 \frac{\sin^2 \beta}{\beta^2}$

$$\frac{dI}{d\beta} = I_0 \left[\frac{2 \sin \beta \cos \beta}{\beta^2} - \frac{2 \sin^2 \beta}{\beta^3} \right] = 0$$

$$\Rightarrow 2 \cos \beta \left[\beta \sin \beta - \frac{\sin^2 \beta}{\cos \beta} \right] = 0$$

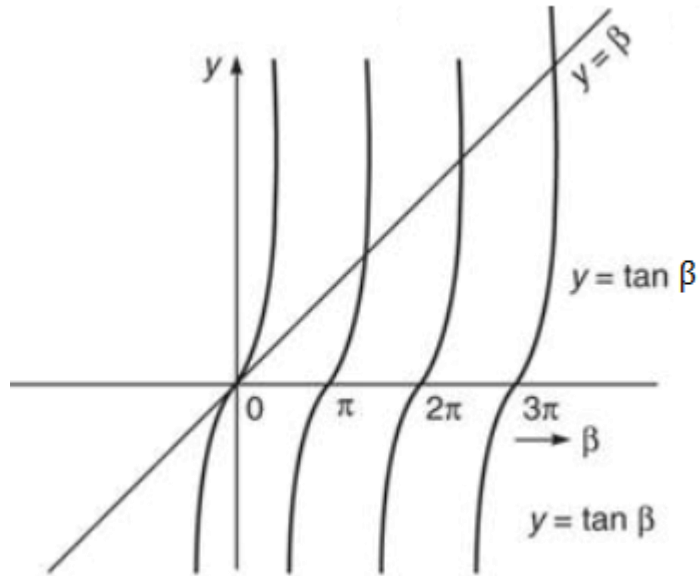
$\Rightarrow$ Either $\cos \beta = 0$ or $\sin \beta [\beta - \tan \beta] = 0$

But $\sin \beta = 0$ corresponds to minima: $\beta = m\pi; m = \phi$

Thus maxima will be given by roots of the transcendental eq:

$$[\beta - \tan \beta] = 0 \text{ i.e. } \tan \beta = \beta$$

Intersections of the plots of $y = \beta$ and $y = \tan \beta$ yield roots i.e. Maxima



$$I = I_0 \frac{\sin^2 1.43\pi}{(1.43\pi)^2}$$

= 0.0496 of the central maximum

$$\beta = 1.43\pi, 2.46\pi, \dots$$

Thanks